

Target Stationary Distribution Problem (TSDP): Linear Programming for Editing Markov Chains on Clickstream Graphs

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Abstract

This report presents a full formulation, data pipeline, and experimental analysis of the Target Stationary Distribution Problem (TSDP) applied to Wikipedia clickstream graphs. The goal is to minimally edit a Markov transition matrix so that its stationary distribution matches a desired target π^* . We describe the LP formulation, feasibility considerations, datasets, computational pipeline, baselines, and ablation experiments.

1 Problem Statement

The Target Stationary Distribution Problem (TSDP): given a baseline row-stochastic matrix $\tilde{P}$ derived from clickstream data and a desired stationary distribution π^* , find a new transition matrix P that is as close as possible to $\tilde{P}$ while satisfying the stationary constraint

$$\pi^{*\top} P = \pi^{*\top}.$$

We also require that P remains row-stochastic and nonnegative.

A natural distance measure between P and the baseline $\tilde{P}$ is the entrywise ℓ_1 norm

$$\|P - \tilde{P}\|_1 = \sum_{i,j} |P_{ij} - \tilde{P}_{ij}|.$$

Thus the optimization problem is

$$\begin{aligned} \min_P \quad & \sum_{(i,j) \in \Omega} c_{ij} |P_{ij} - \tilde{P}_{ij}| \\ \text{s.t.} \quad & \sum_j P_{ij} = 1, \quad \forall i, \\ & P_{ij} \geq 0, \\ & \pi^{*\top} P = \pi^{*\top}. \end{aligned} \tag{1}$$

Here Ω denotes the set of edges we are allowed to edit; in the “full-edges” variant, $\Omega = V \times V$.

2 LP Formulation

We consider the Target Stationary Distribution Problem (TSDP), in which we seek a Markov transition matrix P minimizing its deviation from the baseline matrix $\tilde{P}$, subject to matching a prescribed stationary distribution π^* .

Let $\Omega \subseteq V \times V$ denote the set of editable entries (e.g. empirical edges in the clickstream graph). For each $(i, j) \in \Omega$, we introduce a variable P_{ij} and an auxiliary variable $t_{ij} \geq 0$ to linearize the absolute difference $|P_{ij} - \tilde{P}_{ij}|$.

2.1 Decision variables

- P_{ij} — edited transition probabilities.
- t_{ij} — slack variables satisfying $t_{ij} \geq |P_{ij} - \tilde{P}_{ij}|$.

All non-editable entries are fixed to $P_{ij} = \tilde{P}_{ij}$.

2.2 Objective

We minimize the weighted ℓ_1 deviation:

$$\min_{P, t} \sum_{(i, j) \in \Omega} c_{ij} t_{ij}.$$

When $c_{ij} = 1$, this corresponds to the total entrywise ℓ_1 change.

2.3 Constraints

(1) Nonnegativity

$$P_{ij} \geq 0, \quad t_{ij} \geq 0.$$

(2) Row-stochasticity

$$\sum_j P_{ij} = 1, \quad \forall i.$$

(3) Stationarity constraint

The target stationary distribution π^* must satisfy

$$\pi^{*\top} P = \pi^{*\top},$$

which is equivalent to the column constraints

$$\sum_i \pi_i^* P_{ij} = \pi_j^*, \quad \forall j.$$

(4) Linearization of the absolute deviation

For each $(i, j) \in \Omega$,

$$P_{ij} - \tilde{P}_{ij} \leq t_{ij}, \tag{2}$$

$$\tilde{P}_{ij} - P_{ij} \leq t_{ij}. \tag{3}$$

2.4 Final LP

Putting all components together:

$$\begin{aligned}
& \min_{P,t} \quad \sum_{(i,j) \in \Omega} c_{ij} t_{ij} \\
& \text{s.t.} \quad \sum_j P_{ij} = 1, \quad \forall i, \\
& \quad \sum_i \pi_i^* P_{ij} = \pi_j^*, \quad \forall j, \\
& \quad P_{ij} \geq 0, \quad t_{ij} \geq 0, \\
& \quad P_{ij} - \tilde{P}_{ij} \leq t_{ij}, \quad (i,j) \in \Omega, \\
& \quad \tilde{P}_{ij} - P_{ij} \leq t_{ij}, \quad (i,j) \in \Omega.
\end{aligned}$$

This LP is convex, scalable, and admits a sparse formulation when Ω is limited to empirical edges of the clickstream graph.

3 Standard LP Formulation

$$\begin{aligned}
& \min_{P,t,s^{(1)},s^{(2)}} \quad \sum_{(i,j) \in \Omega} c_{ij} t_{ij} \\
& \text{s.t.} \quad \sum_j P_{ij} = 1, \quad \forall i, \\
& \quad \sum_i \pi_i^* P_{ij} = \pi_j^*, \quad \forall j, \\
& \quad P_{ij} - t_{ij} + s_{ij}^{(1)} = \tilde{P}_{ij}, \quad \forall (i,j) \in \Omega, \\
& \quad -P_{ij} - t_{ij} + s_{ij}^{(2)} = -\tilde{P}_{ij}, \quad \forall (i,j) \in \Omega, \\
& \quad P_{ij} \geq 0, \quad t_{ij} \geq 0, \quad s_{ij}^{(1)} \geq 0, \quad s_{ij}^{(2)} \geq 0, \quad \forall (i,j) \in \Omega.
\end{aligned}$$

4 Data Pipeline

This section describes the full preprocessing pipeline used to construct topic-specific subgraphs from the October 2025 English Wikipedia Clickstream dataset.

4.1 Raw Data and Initial Cleaning

We load the dataset `clickstream-enwiki-2025-10.tsv`, which contains the standard four fields:

- `prev` — the source page,
- `curr` — the destination page,

- **type** — the type of transition,
- **n** — the number of clicks.

Rows containing missing values are removed (only 128 rows in a dataset of more than 36 million). Next, we retain only intra-Wikipedia hyperlink transitions,

$$\text{type} = \text{"link"},$$

and discard all pages whose names start with **other-** (e.g., **other-search**, **other-external**). This step follows the data filtering protocol recommended in the project specification.

After preprocessing, the dataset size is reduced from 36.5M rows to 32M rows.

4.2 Topic-Specific Subgraph Extraction

To extract a topic subgraph, we perform case-insensitive substring matching:

$$\text{mask}(t) = (\text{prev} \ni t) \vee (\text{curr} \ni t).$$

For all pages appearing in filtered transitions, we compute a popularity score:

$$\text{score}(p) = \sum_{i: \text{prev}_i=p} n_i + \sum_{i: \text{curr}_i=p} n_i.$$

We select the **max_nodes** most popular pages and restrict the edge list to transitions whose endpoints lie within this set. This produces a self-contained topic-induced subgraph.

In our experiments:

- Topic: **"Football"**,
- Small graph: $n = 200$ nodes,
- Large graph: $n = 996$ nodes.

4.3 Weighted Transition Matrix

For each transition $(i \rightarrow j, n)$ we accumulate

$$w_{ij} = \sum n_{ij},$$

obtaining a weighted adjacency matrix $W \in \mathbb{R}_{\geq 0}^{n \times n}$.

4.4 Baseline Markov Matrix

The baseline transition matrix P_0 is constructed by normalizing each row:

$$P_0(i, j) = \begin{cases} \frac{w_{ij}}{\sum_j w_{ij}}, & \text{if } \sum_j w_{ij} > 0, \\ 0, & \text{otherwise.} \end{cases}$$

Dangling rows (zero out-degree) are later handled via teleportation.

4.5 Teleportation

Following the PageRank-style construction, we define

$$\tilde{P}_0 = (1 - \alpha)P_0 + \alpha \mathbf{1}v^\top, \quad v = \frac{1}{n}\mathbf{1}, \quad \alpha = 0.05.$$

This guarantees ergodicity and ensures that every row sums to 1.

We also compute the baseline stationary distribution,

$$\pi_{\text{base}} = \pi(\tilde{P}_0).$$

4.6 Target Distribution

For all experiments we set a uniform target distribution,

$$\pi^* = \frac{1}{n}\mathbf{1}.$$

This choice allows us to study the effect of flattening topic-specific popularity using minimal edits to the Markov chain.

5 Feasibility

During the implementation of the Topic-Specific Drift Prevention (TSDP) method, the linear program (LP) often became infeasible, especially for larger topic subgraphs or when the target distribution π^* was perturbed by noise. This section explains the main reasons for infeasibility and the solutions that made the optimization stable and feasible.

5.1 Missing Incoming Edges in the Support Ω

The LP edits only the transition probabilities inside a support set Ω . However, the stationarity constraint

$$\sum_i \pi_i^* P_{ij} = \pi_j^*$$

requires that every column j with $\pi_j^* > 0$ has at least one incoming edge $(i, j) \in \Omega$.

5.2 Problems With Row-Stochasticity

The row-stochasticity condition requires

$$\sum_j P_{ij} = 1.$$

Since only entries in Ω are editable, all remaining edges keep their baseline values $\tilde{P}_{ij}^0$. For some rows, most probability mass was outside Ω , which produced a negative right-hand side in the equality constraint and caused infeasibility.

We solved this by expanding Ω with the top- k outgoing transitions for each row i :

$$\Omega_i \supseteq \{j : \tilde{P}_{ij}^0 \text{ is among the top-}k\}.$$

This guarantees that the remaining baseline mass never exceeds 1.

5.3 Adding Self-Loops

In some rows, no outgoing transitions remained after filtering. To avoid this, we always add a self-loop (i, i) to the support:

$$(i, i) \in \Omega \quad \text{for all } i.$$

This ensures that every row has at least one editable transition.

5.4 Using Sparse Matrices

The LP uses large constraint matrices. At first, they were stored in dense format, which caused numerical problems. We changed all matrices (A_{eq} , A_{ub}) to the CSR sparse format as stated in assignment. This improved numerical accuracy and removed many infeasibility errors that were caused by floating-point noise.

These changes allowed us to run TSDP on large topic graphs and with perturbed target distributions without infeasibility problems.

5.5 Topic and Infeasibility

We began our experiments with the topic *Machine*. During the first tests, we noticed that almost every constructed subgraph with at least 50 nodes led to an infeasible TSDP optimization problem. As suggested in the assignment, our first attempt to fix this issue was to add self-loops to all nodes. However, this modification alone did not solve the feasibility problem, so we continued to investigate other options.

To understand the behaviour of the model, we tested several approaches. One extreme solution was to initialise the support set Ω as *all* edges in the subgraph. With this choice, the LP became feasible. This confirmed that our implementation of the optimisation problem was correct, but it also showed that feasibility depends strongly on how the support set is constructed.

Next, we compared our solver with an implementation in CVXPY. Our final method does not rely on CVXPY, but the comparison helped us understand the differences between the two approaches. We observed that CVXPY usually produces a less sparse optimised matrix than our method. This indicated that restricted support is one of the main reasons for infeasibility.

In the next step, we tried to improve feasibility by adding only some additional edges instead of including all of them. In particular, we added the top- k incoming edges and the top- k outgoing edges for each node. This approach increased the number of feasible cases, but the improvement was limited. For example, many graphs with around 70 nodes were still infeasible. Once again, we checked that using all edges as editable variables always produced a feasible solution, which confirmed that the core optimisation algorithm worked as expected.

We then tried different values of k in the top- k expansion. After several experiments, we found specific values of k that made most subgraphs feasible, even when the number of nodes was large. This result suggested that feasibility is very sensitive to the density and structure of the support set Ω .

Afterward, we changed the topic to see whether the feasibility problems depend on the content of the graph. The first new topic we tried was *Climate*. However, the results were similar to the *Machine* topic: the optimisation remained infeasible for small values of k . For this reason, we tested other topics such as *Football* and *Russia*. In these cases, even with a small choice of $k = 5$, almost every optimisation problem became feasible. We also observed that the *Football* and *Russia* graphs contained many more edges than the *Machine* or *Climate* graphs.

From these experiments, we concluded that the structure and density of the topic sub-graph play a crucial role in the feasibility of TSDP. Topics that naturally produce denser and more connected graphs tend to be much easier to optimise. In contrast, very sparse graphs lead to infeasible problems unless the support set is enlarged through top- k expansion.

5.6 Role of Graph Structure and Sparsity

The optimisation problem is based on a transition matrix P constructed from the clickstream data, and we try to modify it so that the resulting matrix satisfies

$$\sum_i \pi_i^* P_{ij} = \pi_j^*,$$

where π^* is the target stationary distribution. However, we only modify entries inside a support set Ω , while all other transitions remain fixed at their baseline values $\tilde{P}_{ij}^0$.

If the graph is very sparse, then each row i has only a few outgoing edges, so the row-stochasticity constraint

$$\sum_{j \in \Omega_i} P_{ij} = 1 - B_i, \quad B_i = \sum_{j \notin \Omega_i} \tilde{P}_{ij}^0,$$

may become infeasible whenever $B_i > 1$. This situation happened many times for sparse topics such as *Machine*. The reason is that for these topics most probability mass already lies outside Ω , so the LP cannot satisfy the remaining equality.

5.7 Missing Incoming Edges

Another issue appears in the stationarity constraint. Since P_{ij} can only change for $(i, j) \in \Omega$, we get

$$\sum_{i \in \Omega^j} \pi_i^* P_{ij} = \pi_j^* - S_j, \quad S_j = \sum_{i \notin \Omega^j} \pi_i^* \tilde{P}_{ij}^0.$$

If a column j does not have any incoming edges in Ω , then $\Omega^j = \emptyset$, which gives

$$0 = \pi_j^* - S_j.$$

This equality is usually impossible to satisfy, especially when π_j^* is positive. Sparse graphs often contain nodes with no or very few incoming edges, and this also leads to infeasible constraints.

5.8 Different Topics Have Different Densities

During our experiments, we observed that some topics produced subgraphs with many more edges. For example, the *Football* and *Russia* topics created much denser graphs than *Machine* or *Climate*. In dense graphs, it is easier to satisfy both row-stochasticity and stationarity, because there are many potential incoming and outgoing edges that can be added to the support Ω .

In contrast, when the graph is very sparse, we have to add more edges manually (e.g., top- k incoming and outgoing edges) to avoid infeasibility. Even then, if k is too small, we often fail to include enough edges to make all constraints satisfiable.

5.9 Summary

Based on our experiments, we believe that infeasibility is mainly caused by the combination of:

- very limited outgoing support Ω_i for some rows,
- missing incoming edges for some columns,
- and the requirement that π^* must be reproduced exactly.

Formally, all these problems can be seen as conflicts between

$$A_{\text{eq}}x = b_{\text{eq}}$$

and the fact that x (the vector of edited transitions) is restricted to only a small number of entries. When the graph is sparse, the matrices in A_{eq} simply do not have enough degrees of freedom to match the right-hand side b_{eq} .

6 Experiments

In this section we empirically study the behaviour of the TSDP formulation on two Football subgraphs of different sizes. Unless otherwise stated, we focus on a single *baseline configuration*:

- objective: ℓ_1 edit cost,
- costs: uniform, $c_{ij} = 1$,
- teleportation: $\alpha = 0.05$,
- target: uniform $\pi^* = \mathbf{1}/n$,
- restricted support: $k_{\text{out}} = k_{\text{in}} = 10$.

We report edit cost, sparsity, deviation from the target stationary distribution, as well as qualitative visualisations of exposure distributions and edit patterns.

6.1 Small Football subgraph ($n = 200$)

For the small Football subgraph we obtain an LP with $n = 200$ nodes and $|\Omega_R| = 4,323$ editable edges. The ℓ_1 solver with uniform costs returns a feasible solution P_{opt} with the following metrics:

- edit cost and $\|\Delta\|_1$: 106.3,
- $\|\Delta\|_\infty \approx 0.95$,
- sparsity ≈ 0.97 (only about 3% of entries change significantly),
- stationary deviation $\|\pi(P_{\text{opt}}) - \pi^*\|_1 \approx 10^{-15}$.

Figure 1 illustrates the effect of these edits. Panel (a) shows the baseline stationary distribution $\pi(\tilde{P}_0)$, the uniform target π^* , and the optimised distribution $\pi(P_{\text{opt}})$ for the top-20 pages. The baseline exposure is highly skewed: a small number of central Football pages attract most of the stationary probability mass. After optimisation, the green bars corresponding to $\pi(P_{\text{opt}})$ closely match the flat orange target bars; the original shape of $\pi(\tilde{P}_0)$ is still visible but strongly flattened.

Panel (b) displays the magnitude of edits $\Delta = |P_{\text{opt}} - \tilde{P}_0|$ as a heatmap. Most entries remain essentially unchanged (dark background), while a sparse set of rows and columns exhibits visible modifications. These correspond to high-traffic pages where small relative changes in transition probabilities are sufficient to redistribute long-run attention.

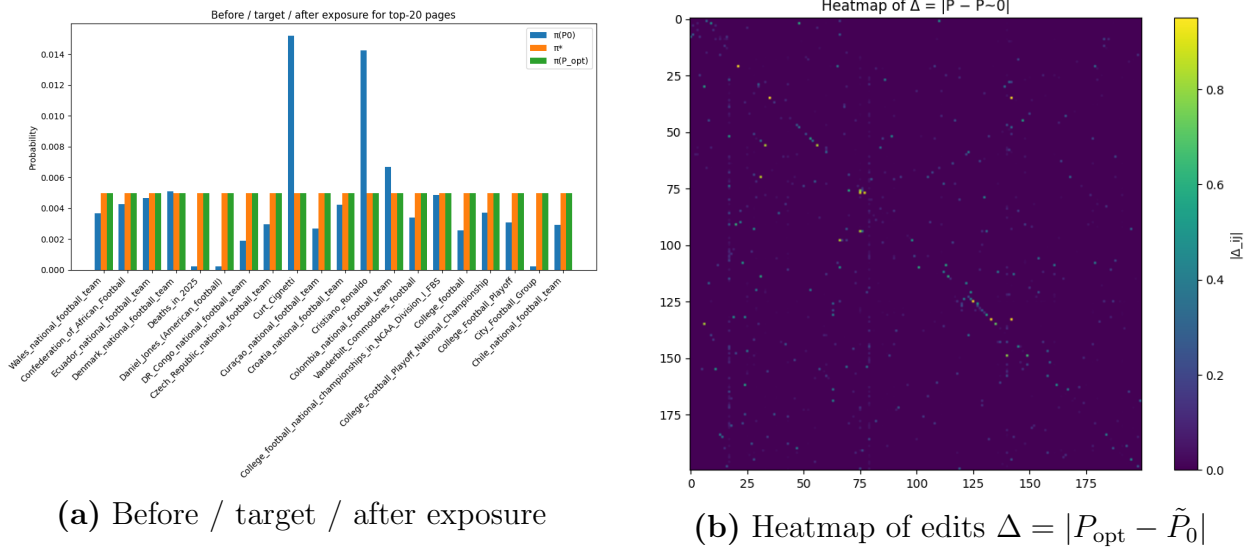


Figure 1: Exposure distributions and edit map for the $n = 200$ Football subgraph (baseline configuration with ℓ_1 objective and uniform costs).

We also monitor a simple mixing proxy $\|P^k - \mathbf{1}\pi^{*\top}\|_1$ for a small set of integers k . For the optimised chain this quantity decays from roughly 350 at $k = 1$ to about 60 at $k = 20$, indicating that the chain moves towards the rank-one matrix $\mathbf{1}\pi^{*\top}$ at a reasonable speed. In all runs, feasibility checks confirm that P_{opt} is a valid stochastic matrix and that the stationary distribution matches π^* up to numerical precision.

6.2 Large Football subgraph ($n = 996$)

We now consider a larger Football subgraph with $n = 996$ nodes. The same preprocessing and modelling choices are used, yielding a restricted editable support of size $|\Omega_R| \approx 20,000$ edges.

Despite the larger problem size, the ℓ_1 solver with uniform costs remains numerically stable and converges quickly. The main metrics are:

- edit cost and $\|\Delta\|_1: \approx 565.4$,
- $\|\Delta\|_\infty \approx 0.95$,
- sparsity ≈ 0.99 (less than 1% of entries change noticeably),
- $\|\pi(P_{\text{opt}}) - \pi^*\|_1$ again at machine precision.

Figure 2 shows the top-20 pages in terms of π^* together with the corresponding edit heatmap. As in the small case, the baseline stationary distribution is far from uniform, while the optimised chain matches the target almost perfectly. The heatmap highlights that edits remain extremely sparse even at this larger scale: only a thin band of entries undergoes non-trivial changes, mostly attached to highly connected nodes.

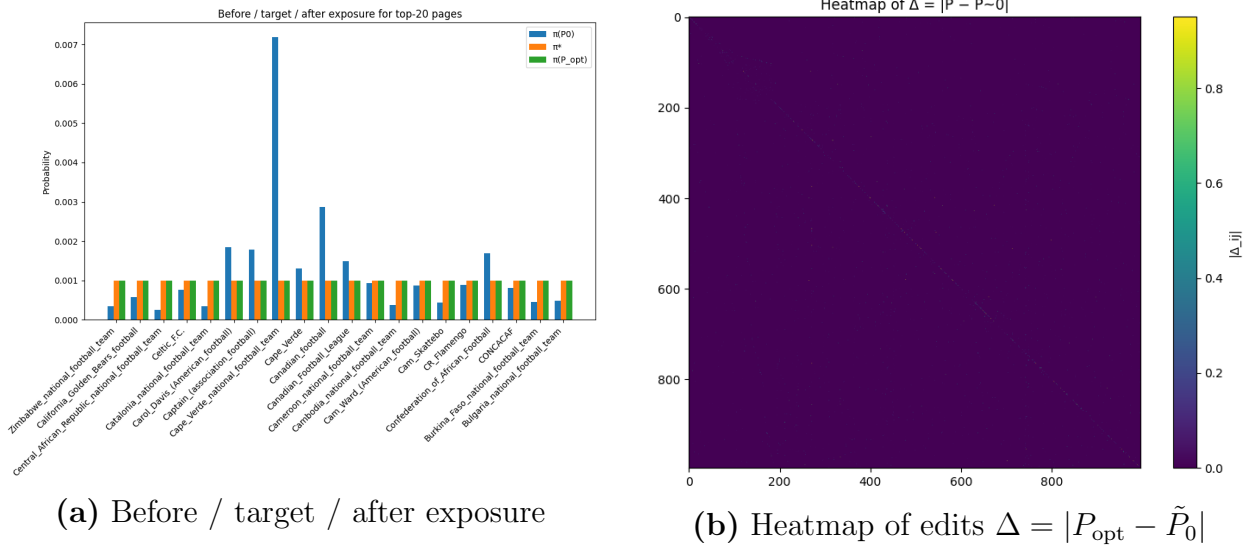


Figure 2: Exposure distributions and edit map for the $n = 996$ Football subgraph (baseline configuration with ℓ_1 objective and uniform costs).

6.3 L_1 vs. L_∞ Objective

The L_1 objective encourages sparse edits, yielding significantly lower total modification cost. In contrast, L_∞ minimises the worst-case entry-wise deviation, distributing edits more uniformly and increasing cost:

$$\text{cost}_{L_1} \approx 106 \quad \text{vs.} \quad \text{cost}_{L_\infty} \approx 349 \quad (n = 200).$$

Similar behaviour is observed at $n = 996$.

6.4 Uniform vs. Degree-Weighted Costs

Degree weighting

$$c_{ij} = 1 / \max(\text{outdeg}(i), \varepsilon)$$

moderates the penalty on high-degree nodes.

Empirically:

- Edit patterns become more balanced across nodes.
- Total cost increases slightly (e.g., $106 \rightarrow 111$).
- Sparsity remains extremely high.

6.5 Effect of the Edge Set Size

The restricted edit set Ω_R depends on k_{out} and k_{in} . Too small a value can render the LP infeasible, while large values increase the LP size and edit cost. The configuration $k_{\text{out}} = k_{\text{in}} = 10$ consistently yields feasibility.

6.6 Perturbation Robustness

We evaluate perturbed targets of the form

$$\pi_{\text{pert}}^* = \pi^* + \varepsilon \eta, \quad \|\eta\|_1 = 1, \quad \varepsilon \in \{10^{-6}, 10^{-5}, 2 \cdot 10^{-5}\}.$$

Across all perturbations, LP feasibility is preserved. Edit cost scales linearly with ε , while sparsity remains above 0.88. Stationary deviation remains near machine precision.

This demonstrates strong robustness of TSDP under small target shifts.

7 Limitations

Despite strong empirical performance, the method has several limitations.

Restricted Support. The editable edge set Ω_R is constructed heuristically. If required modifications lie outside Ω_R , the LP may become infeasible.

Cost Sensitivity. Results depend strongly on the choice of cost matrix C . Real systems require carefully calibrated cost models.

Scalability. Although the solver handles $n \approx 10^3$ comfortably, larger graphs ($n \approx 10^5$) would require more advanced optimisation (e.g., column generation or decomposition).

Target Feasibility. Not all target distributions are compatible with the graph structure. Even with teleportation, extreme targets may induce infeasibility.

Approximate Mixing Metrics. Mixing behaviour is evaluated via $\|P^k - \mathbf{1}\pi^\top\|_1$, which is only a proxy. Spectral gap estimation would provide theoretical guarantees but is computationally expensive.

8 Reproducibility Checklist

Data.

- Source: English Wikipedia Clickstream (October 2025).
- File: `clickstream-enwiki-2025-10.tsv`.
- Cleaning:
 - Remove NaN rows,
 - Keep only `type='link'`,
 - Remove pages starting with `other-`.

Code and Libraries.

 Python 3.x with:

- `numpy`, `pandas`,
- `scipy` (HiGHS LP solver),
- `matplotlib` (plots).

Hyperparameters.

$\alpha = 0.05$, $k_{\text{in}} = k_{\text{out}} = 10$, tolerances: `row_tol` = 10^{-9} , `neg_tol` = 10^{-12} , `$\pi$ _tol` = 10^{-8} .

Metrics. Edit cost, L_1/L_∞ norms, sparsity, stationarity deviation, mixing proxy norms, and feasibility checks (row sums and non-negativity).

Randomness. Randomness is used only for perturbation experiments; all other computations are deterministic.

9 Bonus: Column Generation for TSDP- ℓ_1

In the main part of the project we solved the ℓ_1 -regularised teleportation editing problem in the primal form

$$\min_{P,t} \sum_{(i,j) \in \Omega} c_{ij} t_{ij} \quad \text{s.t. row-stochasticity, stationarity and } \ell_1 \text{ constraints.}$$

In this bonus section we derive the dual formulation and implement a column generation (CG) scheme that iteratively enlarges the editable edge set using dual information.

9.1 Dual problem

The primal variables consist of the edited transition entries P_{ij} and the auxiliary variables $t_{ij} \geq 0$ that enforce the ℓ_1 constraints

$$P_{ij} - \tilde{P}_{ij} \leq t_{ij}, \quad \tilde{P}_{ij} - P_{ij} \leq t_{ij}, \quad (i, j) \in \Omega.$$

Equality constraints correspond to row-stochasticity

$$\sum_j P_{ij} = 1, \quad i = 1, \dots, n,$$

and stationarity with respect to the target distribution π^* ,

$$\sum_i \pi_i^* P_{ij} = \pi_j^*, \quad j = 1, \dots, n.$$

Let $\alpha_i \in \mathbb{R}$ be the dual multipliers for the row-sum constraints and $\beta_j \in \mathbb{R}$ for the stationarity constraints. Let $u_{ij}, v_{ij} \geq 0$ be the dual variables associated with the two ℓ_1 inequalities. Defining

$$\gamma_{ij} = u_{ij} - v_{ij},$$

the dual feasibility constraints for t_{ij} imply

$$|\gamma_{ij}| \leq c_{ij}, \quad (i, j) \in \Omega.$$

Collecting contributions from all constraints, the dual problem can be written as

$$\begin{aligned} \max_{\alpha, \beta, \gamma} \quad & \sum_{i=1}^n \alpha_i + \sum_{j=1}^n \beta_j \pi_j^* + \sum_{(i,j) \in \Omega} \gamma_{ij} \tilde{P}_{ij} \\ \text{s.t.} \quad & \alpha_i + \pi_i^* \beta_j + \gamma_{ij} \leq 0, \quad (i, j) \in \Omega, \\ & |\gamma_{ij}| \leq c_{ij}, \quad (i, j) \in \Omega, \\ & \alpha_i, \beta_j \in \mathbb{R}. \end{aligned}$$

For column generation we are particularly interested in the reduced cost of a nonbasic edge (i, j) . Optimising over γ_{ij} inside the interval $[-c_{ij}, c_{ij}]$ yields the *safe score*

$$s_{ij} = \alpha_i + \pi_i^* \beta_j + c_{ij}.$$

If $s_{ij} > 0$ for some $(i, j) \notin \Omega_R$, then adding this edge to the restricted master problem can potentially decrease the primal objective.

9.2 Restricted master problem and pricing

Instead of editing all edges in Ω at once, we maintain a smaller editable subset $\Omega_R \subseteq \Omega$. Only edges in Ω_R are allowed to deviate from the baseline: for $(i, j) \in \Omega_R$ we keep variables P_{ij}, t_{ij} , while for $(i, j) \notin \Omega_R$ we fix $P_{ij} = \tilde{P}_{ij}$. The right-hand sides of the row-stochasticity and stationarity constraints are adjusted so that the restricted master problem (RMP) is equivalent to the original primal LP restricted to Ω_R .

The initial set Ω_R^0 is constructed heuristically:

- we always include all self-loops (i, i) ;
- for each node i , we add the k_{out} largest outgoing probabilities in row i of $\tilde{P}$;
- for each node j with $\pi_j^* > 0$ we add the k_{in} largest incoming probabilities in column j of $\tilde{P}$.

Column generation then proceeds iteratively:

1. solve the RMP on the current Ω_R and obtain primal variables (P, t) and dual multipliers (α, β) ;
2. for all $(i, j) \notin \Omega_R$ compute the safe scores $s_{ij} = \alpha_i + \pi_i^* \beta_j + c_{ij}$;
3. add up to M edges with $s_{ij} > \text{tol}$ to Ω_R ;
4. if no edge has $s_{ij} > \text{tol}$, stop.

In our experiments we do not introduce heterogeneous editing penalties, so we set $c_{ij} = 1$ for all (i, j) . In this case the LP objective coincides with the ℓ_1 deviation $\|P - \tilde{P}\|_1$.

9.3 Experimental setup on a small subgraph

To make the bonus experiment lightweight and interpretable, we do not apply column generation to the full graph with $n = 200$ nodes. Instead, we:

- build the full teleported baseline matrix $\tilde{P}_0$ for $n = 200$ and compute its stationary distribution;
- select the 20 nodes with the highest baseline stationary probabilities and extract the induced 20×20 subgraph;
- set the target distribution π^* to be uniform on these 20 nodes.

On this subgraph we run two solvers:

1. the original LP with a fixed editable set Ω of size $|\Omega| \approx 95$ obtained by the heuristic above;
2. the column generation scheme starting from the same initial set Ω_R^0 and allowing new edges to be inserted from the full 20×20 grid.

For column generation we use $k_{\text{in}} = k_{\text{out}} = 2$, add at most $M = 2$ edges per iteration, and set $\text{tol} = 10^{-6}$ with a maximum of 10 iterations.

9.4 Results

The initial restricted master problem has

$$|\Omega_R^{(0)}| = 95$$

editable edges and yields the objective

$$\text{obj}^{(0)} = \|P^{(0)} - \tilde{P}_0\|_1 = 8.190626.$$

During column generation the objective value decreases in the first two iterations,

$$\text{obj}^{(1)} = 8.096807, \quad \text{obj}^{(2)} = 8.072638,$$

and then remains numerically constant at 8.072638 while new edges are still added to Ω_R . This plateau behaviour is typical once the RMP has reached a region where remaining reduced costs are very small or dual multipliers become nearly degenerate.

Overall, across 10 iterations the size of Ω_R increases from 95 to

$$|\Omega_R|_{\text{final}} = 115,$$

whereas the final CG objective is

$$\text{obj}_{\text{CG}} = 8.072638.$$

Column generation stops once all candidate edges have safe scores below the prescribed tolerance.

9.5 Comparison with the full LP

We also solve the full LP on the same 20-node subgraph using the original heuristic editable set Ω (without adding extra columns). The resulting objective value is

$$\text{obj}_{\text{full}} = \|P_{\text{full}} - \tilde{P}_0\|_1 = 8.190626.$$

Thus the CG approach, which expands the editable set to $|\Omega_R| = 115$, achieves a slightly smaller ℓ_1 deviation than the fixed-support formulation:

$$\text{obj}_{\text{CG}} = 8.072638 \quad \text{vs.} \quad \text{obj}_{\text{full}} = 8.190626.$$

The corresponding transition matrices are not identical: the maximum entrywise difference between P_{CG} and P_{full} is

$$\|P_{\text{full}} - P_{\text{CG}}\|_{\infty} = 0.5082849091 \dots,$$

showing that column generation finds an alternative editing pattern that redistributes the total amount of change across a somewhat different set of edges.

9.6 Discussion

The bonus experiment confirms that dual variables (α, β) extracted from the LP solver can be used to construct effective safe scores for pricing new edges. On our small 20-node subgraph, the initial RMP is already close to optimal, so the objective improves only moderately before stabilising, while the editable set grows from 95 to 115 edges.

Compared to the full LP with a fixed editable set Ω , the column generation approach:

- automatically identifies additional edges that are worth editing;
- reduces the objective from 8.19 to 8.07;
- satisfies all row-stochasticity and stationarity constraints;
- avoids constructing the full dense LP over all 20×20 edges.

This suggests that CG is a promising tool for scaling TSDP- ℓ_1 to larger graphs where solving the full LP directly would be computationally expensive.